

## Dr. Dogga Raveendhra

Ex-Senior Engineering Consultant, **ALTEN GmbH, Germany**  
Post-Doc (**Renewable Energy Research Centre, AUS**)  
Visiting Researcher (**North Carolina State University, USA**)  
Ph.D. (IIT Roorkee, **Excellence in Doctoral Research Awardee**)  
M.Tech. (Electrical Drives and Power Electronics, **EED, IITR**)

Hard-working, results-oriented and self-driven researcher with 8 years of experience in Power Electronics and Electrical Drives. Proven history in overseeing all aspects of projects from initial planning, developing, prototyping, and publishing. Experience accompanied by graduate degree, teaching, and research with the best innovation award. Possess strong communication, team management, negotiation, leadership, training, problem-solving and motivational skills.



**Assistant Professor (Gr-I), EED, NITW**

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## INTERESTS

1. Applications and Controls of Power Electronics in Automotive/Transportation, and Renewable Energy Systems
2. Dynamic Modelling and Control of Power Electronic Converters and Motor-Drives
3. Applications of Artificial Intelligence in Power Systems and Power Electronics
4. Wide Band-Gap Devices based Power Converters
5. FPGA and DSP Controllers for PE Applications

## Profile Summary

1. **Excellence in Doctoral Research Award** from IIT Roorkee for best Ph. D. Work
2. Ph.D. (Thesis grade "A") from EE department, **IIT Roorkee**.
3. **Best Innovation** award by Zunik Energies in 2019.
4. Countries travelled on research Purpose: **USA, Germany, Malaysia, and UAE**
5. **Patents Filed/Published:** 01 US Patent Disclosures and 02 published Indian Patents
6. **10+ years of Industrial R&D and Academic experience** in various engineering domains
7. Worked as Visiting Researcher at **FREEDM System Centre, NC State University, USA**. Involved in the Development of Novel Characterization Circuit for Power Electronics Devices Testing.
8. Worked as an **Assistant Professor** at the Electrical Engineering Department in Graphic Era University, Dehradun.
9. Worked in tools as Eagle software, Altium (for PCB design), Xilinx ISE for FPGA Controllers Spartsan-6 and 3E; familiar with TI-DSP processor - TMS320F240, TI - Processor - TMS320F148, MSP-430.

## Few outcomes from PhD Work apart from Publications/Grants:

10. Nominated by Director IIT Roorkee for Festival of Innovation and Entrepreneurship (FINE-2018), Rashtrapati Bhavan, New Delhi.
11. Nominated by TIDES Business Incubator, IIT Roorkee for the exhibition at Destination Uttarakhand (UKIS-201) (Chief Guests are Prime Minister of India and CM of Uttarakhand)
12. Invited delegate for CII DST India – Italy Technology Summit held on 29<sup>th</sup> and 30<sup>th</sup> Oct, 2018 at Taj Palace, New Delhi (Chief Guest are Prime Minister of India and Prime Minister of Italy).
13. Invited to showcase Innovation at FICCI 14<sup>th</sup> Higher Education Summit
14. Selected for SLUSH 2018, which was held on 03<sup>rd</sup> to 05<sup>th</sup> December 2018 in Helsinki, Finland. (Funded by AP Innovation Society and Invest India)

## HIGHLIGHTS OF QUALIFICATIONS

1. Design, modelling, simulation, control, and implementation of power electronic converters and machine drive, including space vector inverters for automotive/transportation and industrial motor drives, and single-stage boost-inverters for renewable energy applications
2. Simulation (Model-Based Analysis) in Simulink, including modelling and control of DC-AC and DC-DC power electronic converters in continuous and discrete domains
3. Programming in MATLAB, including numerical analysis methods and artificial intelligence algorithms
4. Technical support to world-leading OEMs, including writing datasheets, application notes, and training materials, presentations, schematic and PCB layout review, and failure analysis.
5. Testing and troubleshooting of DC-AC and DC-DC power electronic converters
6. Using laboratory test equipment, including oscilloscopes, dynamometer, DC and AC power supplies, source meters, load emulators, power analysers, RLC meters, signal generators, thermal camera, and logic analyser

## Educational Qualifications

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| <b>Ph.D.</b>  | Thesis: Improved Power Quality Transformerless Power Converters for Solar PV applications<br>Electrical Engineering Department, <b>Indian Institute of Technology, Roorkee</b>             | <b>2019 with A grade</b> |
| <b>M.Tech</b> | Dissertation: Development of Converter System for Solar PV Power Generation<br>Specialization of Electrical Drives and Power Electronics<br><b>Indian Institute of Technology, Roorkee</b> | <b>2012 with 88.33%</b>  |

### International Work Experience

**Senior Engineering Consultant**, ALTEN GmbH, Dortmund, North Rhine-Westphalia, Germany from **Dec'22-Till June' 2023**

**Roles and Responsibilities:**

- Develop Controller algorithms for on-board chargers for Electric Vehicles.
- Develop, test, and implement innovative research for power converters of Electric Vehicle Systems.

**Post-Doctoral Research Associate**, Renewable Energy Research Center, American University of Sharjah, UAE as from **Sep'19-June'20**  
(1 Academic Year)

**Roles and Responsibilities:**

- Develop, test, and implement innovative research for Hybrid Energy Storage Systems of Electric Vehicle Systems.
- Develop prototypes & tools for Hybrid Energy Storage Systems of Electric Vehicles.
- Development of a loss-thermal model for space vector modulated (SVM) inverters, used for IGBT/diode sizing
- Ripple analysis and capacitor sizing of traction/automotive inverters
- Fault and transient analyses of the inverters
- Development, simulation, and experimental verification of a field-oriented control for a 400V IPM traction/automotive drive

**Major Projects Handled**

- LC-Bidirectional DC-DC Converter, Super Capacitor Characterisation using SiC Converters, Elimination of Common Mode Voltages in power Converters and Wireless Power Supply for Drones

**Visiting Doctoral Researcher, FREEDM Systems Centre, NC State University, USA** from **July'16 to Nov'16**

**Roles and Responsibilities:**

- Design and Development of Power Converter for Solar PV Systems.

**Major Projects Handled**

- Novel Characterization Circuit for Power Semiconductor Device Testing
- Design and verification of an IGBT gate drive circuit
- Design, tuning, and verification of IGBT short circuit (DESAT) and thermal protection
- Design, simulation, documentation, and experimental validation of a sliding mode controller for Solar Powered Inverters
- Development, implementation, and documentation of a DC bus voltage prediction method to improve THD
- Design and implementation of an input phase open fault detection scheme for reduced DC capacitor drives

### Work Experience in India

**Assistant Professor, MNNIT Allahabad**, India from **June'26 to Till Date**

**Roles and Responsibilities:**

- Teaching, Research & Development
- Member, Development Coordination Committee (DCC) Sub-Committee of Quantum Technology Division in TEC, Govt. of India (**National Level**).

**Major Projects Handling:**

- GaN based LED Drivers, On-board Chargers for Electric Vehicles, LLC Resonant DC-DC Converters and Power Conditioning Units for Solar PV Power Generations

**Assistant Professor, MNNIT Allahabad**, India from **June'23 to June'26**

**Roles and Responsibilities:**

- Teaching, Research & Development
- Member, Development Coordination Committee (DCC) Sub-Committee of Quantum Technology Division in TEC, Govt. of India (**National Level**).
- Convener, Institute Committee for Center for Sustainable Energy (**Institute Level**)
- Associate Faculty In-Charge [Students Counselling (Boys)] (**Institute Level**)
- Coordinator, Website Management and Database Committee, EED (**Department Level**)
- O.C. (Officer-in-Charge) for Power Electronics Lab, EED (**Department Level**)
- Member, DUGC (EED) (**Department Level**)
- Member, DMPC (EED) (**Department Level**)
- Department Doctoral Selection Committee, EED, MNNIT (**Department Level**)
- Faculty Representative (EED) for Indian Science Technology and Engineering facilities Map (I-STEM), an initiative of Govt. of India, under the PM-STIAC mission

**Major Projects Handling:**

- On-board Chargers for Electric Vehicles, LLC Resonant DC-DC Converters and Power Conditioning Units for Solar PV Power Generations

**Senior Research Engineer/Tech Lead/Founder, Zunik Energies Pvt. Ltd., India as from Nov'17**

**Roles and Responsibilities**

- Research and product development of Power Electronic Converters for Renewable Energy Applications and Electrical Vehicles.
- Maintain current knowledge of technology landscape and developments
- Track, analyse and monitor technology performance metrics and consolidate our technology platforms and create plans for each
- Identify new areas of opportunities and create plans to implement them for our clients and us
- Develop, test, and implement processes for product manufacture and production.
- Developed prototypes & tools and leveraged current applications
- Evaluating and implementing new systems and infrastructure
- Discover and implement new technologies that yield competitive advantage
- Preparing research proposals and funding applications

**Major Projects Handled**

- Single Stage Converters for Solar PV Applications, High Gain Converters for Electrical Vehicles and FPGA Controlled Power Converters

**Products Developed**

- Single Stage Capacitor Clamped Boost Inverter, Super Capacitor Characterization Circuit using SiC-based Power Converter, 200kW SiC-based Central Inverter, LC-Bi-Directional DC-DC Converter for Solar Applications.

**Associate Professor, GRIET Hyderabad, India from Oct'20 to Nov'22**

**Roles and Responsibility:**

- Teaching, Research & Development
- Conduct Research in Industrial Drives Lab
- M.Tech Coordinator

**Major Projects Handling:**

- Capacitor Clamped Boost Inverter for on-grid Solar PV Applications, MSME, Govt. of. India
- Design of low cost and highly efficient testing circuit for power semiconductor device characterization, DST, Govt. of. India

**Senior Research Fellow, IIT Roorkee, India from June'16 to July'19 (3 Years)**

- Design, dynamic modelling and control, simulation, and implementation of MOSFET and IGBT based buck, boost, and single-stage boost-inverter
- Development and experimental verification of a phasor-based PWM switching scheme for single- and three-phase single-stage boost-inverters
- Simulation and Hardware Implementation of Power Electronics Converters for Solar PV Applications

**Junior Research Fellow, IIT Roorkee, India from July'14 to May'16 (2 Years)**

**Roles and Responsibilities**

- Worked on Capacitor Clamped Boost Inverter Project Funded by TIDES, IIT Roorkee

**Major Projects Handled**

- Design and Development of Power Converter for Solar PV Systems.
- Modelling, simulation, and experimental validation of photovoltaic (PV) systems
- Design, dynamic modelling and control, simulation, and implementation of MOSFET and IGBT based buck, boost, and single-stage boost-inverter
- Development and experimental verification of a phasor-based PWM switching scheme for single and three-phase single-stage boost- inverters

**Assistant Professor, Graphic Era University, India from July'12 to July'14 (2 years)**

**Roles and Responsibility:**

- Teaching and Research
- Conduct Research in Industrial Drives Lab
- Faculty advisor/course coordinator for Advanced Control and Industrial Drives Lab
- Working on Developing Supervisory Controls for Hybrid Electric Vehicles

**Major Projects Handled**

- Multi-Level Converter and Control Using FPGA for Drives Applications and Bi-Directional Power Converters for Solar PV Applications

**Publications (Journals, Patents, Book Chapters and Conferences): Citations: 605, h-index: 14, i10-index: 15**

#### Journals Disclosures

1. D. Raveendhra and M. K. Pathak, "Three-Phase Capacitor Clamped Boost Inverter," in *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 7, no. 3, pp. 1999-2011, 2018, doi: 10.1109/JESTPE.2018.2873154, Print ISSN: 2168-6777, Electronic ISSN: 2168-6785 (Impact Factor: 5.177). (SCIE Indexed, Q1 Journal)
2. Raveendhra Dogga, M.K. Pathak, "Recent trends in solar PV inverter topologies", *Solar Energy (Elsevier)*, Volume 183, 2019, Pages 57-73, ISSN 0038-092X, doi: 10.1016/j.solener.2019.02.065 (Impact Factor: 4.608). (SCIE Indexed, Q1 Journal)
3. S. Mukhopadhyay, R. Dhaouadi, M. Takroui and R. Dogga, "Supercapacitor Characterization Using Universal Adaptive Stabilization and Optimization," in *IEEE Open Journal of the Industrial Electronics Society*, vol. 1, pp. 166-183, 2020, doi: 10.1109/OJIES.2020.3008339. (Impact Factor: 5.2). (ESCI Indexed, Q1 Journal)
4. D Raveendhra \*, Poojitha Rajana, BL Narasimharaju, Babu Y Suri, Eugen Rusu\*, Fayek H Hady. Analysis and Operation of High DC-AC Gain 3- $\phi$  Capacitor Clamped Boost Inverter. *Energies* 2022, 15(8), 2955; <https://doi.org/10.3390/en15082955> (Impact Factor: 3) (SCIE Indexed, Q1 Journal)
5. Dogga Raveendhra; Praveen Jugge; Ravi Kumar KS; Poojitha R; Denis Sidorov, Daniil Panasetzky; Narasimha Raju BL, "Part I: State-of-the-Art technologies of DC Micro Grid with Hybrid Energy Storage Systems: Architectures and Converter Topologies", *Energies*, <https://doi.org/10.3390/en16020923> (Impact Factor: 3) (SCIE Indexed, Q1 Journal))
6. B. L. Narasimharaju, U. Ramanjaneya Reddy and R. Dogga, "Design and analysis of voltage clamped bidirectional DC–DC converter for energy storage applications," in *The Journal of Engineering*, vol. 2018, no. 7, pp. 367-374, 7 2018, doi: 10.1049/joe.2018.0127. Open Access, Online ISSN 2051-3305 (ESCI Indexed, Q2 Journal)
7. Dogga Raveendhra, Rached Dhaouadi, Habib-ur Rehman, Shayok Mukhopadhyay, "LC Impedance Source Bi-Directional Converter with Reduced Capacitor Voltages", *Electronics* 2020, 9(7), 1062; <https://doi.org/10.3390/electronics9071062>. Open Access (Impact Factor: 2.397) (SCIE Indexed, Q2 Journal)
8. Kumar, K.S.R.; Pandian, A.; Sastry, V.V.; Raveendhra, D. Capacitor Clamped Coupled Inductor Bi-Directional DC-DC Converter with Smooth Starting. *Machines* 2022, 10, 47. <https://doi.org/10.3390/machines10010047> Open Access (Impact Factor: 2.428) (SCIE Indexed, Q2 Journal)
9. Dogga Raveendhra, Poojitha Rajana, Ravi Kumar KS, Praveen Jugge, Ramesh Devarapalli, Eugen Rusu, Hady H. Fayek, "High Gain Multi Phase Interleaved Differential Capacitor Clamped Boost Converter", *Electronics* 2022, 11(2), 264; <https://doi.org/10.3390/electronics11020264> Open Access (Impact Factor: 2.397) (SCIE Indexed, Q2 Journal)
10. Rao, D. S. N. M., Raveendhra, D., Kumar, D. G., Narukullapati, B. K., Rao, D. S., & Palakaluri, S. D. (2021). Comparison Investigation into Power System Optimization and Constraint-Based Generator Load Scheduling Using Metaheuristic Algorithms. *ECTI Transactions on Electrical Engineering, Electronics, and Communications*, 19(2), 200-208. (SCOPUS Indexed, Q3 Journal)
11. Satyesh, P. Ravikanth, D. Raveendhra, "Rectifier Load Analysis for Electric Vehicle Wireless Charging System using Fuzzy Logic Controller", *High Technology Letters*, Vol. 27 (7), 2021. ISSN NO: 1006-6748 (Q3 SCOPUS Journal)
12. Vadlakonda Ranadeep, Dogga Raveendhra, "Control of Capacitively Coupled Impedance source type Wireless Power Transfer Network ", *High Technology Letters*, Vol. 27 (7), 2021. ISSN NO: 1006-6748 (Q3 SCOPUS Journal)
13. Keerthi Volisetty, Dola Gobindha Padhan, Dogga Raveendhra, "Single Phase H Bridge Inverters Without Using Additional Power Electronics by Power Decoupling Strategy ", *High Technology Letters*, Vol. 27 (10), 2021. ISSN NO: 1006-6748 (Q3 SCOPUS Journal) (<https://doi.org/10.37896/HTL27.10/4465>)
14. Raveendhra, D.; Poojitha, et. al., Part II: State-of-the-Art Technologies of Solar-Powered DC Microgrid with Hybrid Energy Storage Systems: Converter Topologies. *Energies* 2023, 16, 6194. <https://doi.org/10.3390/en16176194>. (Impact Factor: 3) (SCIE Indexed, Q1 Journal)
15. Raveendhra, D., Lavanya, K., Ravi Kumar, K.S., Koreboina, V.B., Pakalapati, J. and Gopal, Y. (2025), Impact of Parasitics on the Dynamic Performance of Capacitor Clamped Bidirectional DC–DC Converter. *IET Power Electron.*, 18: e70059. <https://doi.org/10.1049/pel2.70059> [SCIE Journal, Impact Factor: 2]
16. G. Nagesh, D. Raveendhra et.al, "Extraction and Energy Management of Solar Photovoltaic, Fuel Cell, and Battery Systems Using a Novel Multi Port Power Converter for DC Microgrid Applications," in *IEEE Access*, vol. 13, pp. 87418-87437, 2025, doi: 10.1109/ACCESS.2025.3570190. [SCIE Journal, Impact Factor: 3]
17. G. Nagesh, D. Raveendhra, et.al "Integration of Renewable Energy Sources with Energy Storage Devices Using a Novel Four Port Converter for DC Microgrid Applications," *Engineering Research Express*, DOI: 10.1088/2631-8695/ade224 [ESCI Journal, Impact Factor: 1.5].
18. Guguloth Nagesh, Dogga Raveendhra, et.al.; "High Voltage Gain Transformer-Less DC-DC Converter for Fuel Cell Electric Vehicles", *IEEE Open Journal of the Industrial Electronics Society*. (Impact Factor: 4.3) (Accepted). (ESCI Indexed, Q1 Journal)
19. Guguloth, Nagesh, Dogga Raveendhra, et.al.; "Four Port Non-Isolated DC-DC Converter for Fuel-Cell Electric Vehicles" *IEEE Transactions on Transportation Electrification* (Under Review)

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| 20. Guguloth, Nagesh, Dogga, Raveendhra, et.al.; "Operation and Analysis of a Four Port Non-Isolated DC-DC Converter for Energy Storage Systems in DC Micro-Grid Applications" Journal of Energy Storage (Under Review)                                                                                                                                                                                                                    |
| 21. Dogga Raveendhra, Douglas C Hopkins, Utkarsh Mehrotra, Hady H. Fayek, "Neutralized CMV Inverter (NCI) for Electrical Vehicle Applications" IEEE Power Electronics (SCIE Indexed) (In Communication)                                                                                                                                                                                                                                    |
| <b>Patent Disclosures (Granted: 1, Published: 3)</b>                                                                                                                                                                                                                                                                                                                                                                                       |
| 22. Raveendhra Dogga, Ravi Kumar KS, A Pandian, V. V. Sastry Vedula, Poojitha Rajana, Praveen Jugge, "High Gain Non-Isolated Bi-Directional DC-DC Converter" <b>Granted</b> on 09th November, 2021 (Patent No: 418205)                                                                                                                                                                                                                     |
| 23. <b>US Patent Disclosure:</b> Douglas C Hopkins, Dogga Raveendhra, "Energy Re-Circulation Circuit and Controls for Applications including Power Semiconductor Device Characterisation", Filed at Office of Technology Commercialisation and New Ventures, NCSU for US patent. Ref. No: 17126                                                                                                                                            |
| 24. <b>Indian Patent:</b> Raveendhra Dogga, M K Pathak, "Novel Single-Stage Inverter", Application Number: 201641038706, <b>Published</b>                                                                                                                                                                                                                                                                                                  |
| 25. <b>Indian Patent:</b> Ravi Kumar KS, Raveendhra Dogga, Nagesh Kumar, V V Sastry Vedula, Novel Capacitor Clamped Bidirectional DC-DC Converter, Application Number: 201641038706, <b>Published</b>                                                                                                                                                                                                                                      |
| <b>Book Chapters (6)</b>                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 26. Dogga Raveendhra, M Prashanth, K Sudha, "Effects of Common Mode Voltage in ZSI based Induction Motor Drive for EV applications", 3 <sup>rd</sup> International Conference on Machine Learning, Advances in Computing, Renewable Energy and Communication, 10-11 December 2021.                                                                                                                                                         |
| 27. Dogga Raveendhra, K Sudha, M Prashanth, "Single Stage Power Conditioning Unit for Battery Assisted, Solar Powered Remote Area Power Supply", 3 <sup>rd</sup> International Conference on Machine Learning, Advances in Computing, Renewable Energy and Communication, 10-11 December 2021.                                                                                                                                             |
| 28. Raveendhra D, Thakur P., Chauhan A. (2015), FPGA Controlled Power Conditioning System for Solar PV Fed PMDC Motor. In: Kamalakannan C., Suresh L., Dash S., Panigrahi B. (eds) Power Electronics and Renewable Energy Systems. Lecture Notes in Electrical Engineering, vol 326. Springer, New Delhi, Online ISBN: 978-81-322-2119-7 ( <i>Indexing: ISI Proceedings, EI-Compendex, SCOPUS, EI Compendex, MetaPress, Springerlink</i> ) |
| 29. Chauhan A., Thakur P., Raveendhra D. (2015) Quantification of Voltage Unbalance Conditions. In: Kamalakannan C., Suresh L., Dash S., Panigrahi B. (eds) Power Electronics and Renewable Energy Systems. Lecture Notes in Electrical Engineering, vol 326. Springer, New Delhi, Online ISBN: 978-81-322-2119-7. ( <i>Indexing: ISI Proceedings, EI-Compendex, SCOPUS, EI Compendex, MetaPress, Springerlink</i> )                       |
| 30. DBLP Lecture Notes in Computer Science Engineering: Raveendhra, Dogga, "Simulation-based study of FPGA based controller for Single-Phase Matrix Converter for different types of loads", IDEAS CPS in LSCS Series. ( <i>Indexing: Thomson ISI Proceedings, DBLP, IET Inspec, Scopus, EI Compdex, Google Scholar, ProQuest, etc.</i> )                                                                                                  |
| 31. Shiv Prakash Singh, Suneel Raju Pendem, Brij Kishor, Dogga Raveendhra and M Venkatesh Naik, "A Review on Hybridization and Energy Management Strategies for Hydrogen based Fuel Cell Electric Vehicle Applications," 1st Edition of Hydrogen Energy Production, Storage and Utilization, Taylor & Francis.                                                                                                                             |
| <b>Conference Proceedings (37)</b>                                                                                                                                                                                                                                                                                                                                                                                                         |
| 32. D. Raveendhra, M. K. Pathak and A. Panda, "Power conditioning system for solar power applications: Closed loop DC-DC converter fed FPGA controlled diode clamped multilevel inverter," 2012 IEEE Students' Conference on Electrical, Electronics and Computer Science, Bhopal, 2012, pp. 1-4, doi: 10.1109/SCECS.2012.6184820. ( <i>Indexed in SCOPUS</i> )                                                                            |
| 33. D. Raveendhra, K. P. Guruswamy and P. Thakur, "FPGA based 2-stage power conditioning system for PV power generation," 2013 International Conference on Power, Energy and Control (ICPEC), Sri Rangalatchum Dindigul, 2013, pp. 44-50, doi: 10.1109/ICPEC.2013.6527622. ( <i>EI indexed and Indexed in SCOPUS</i> )                                                                                                                     |
| 34. S. Semwal, D. Joshi, R. S. Prasad and D. Raveendhra, "The practicability of ICA in home appliances load profile separation using current signature: A preliminary study," 2013 International Conference on Power, Energy and Control (ICPEC), Sri Rangalatchum, Dindigul, 2013, pp. 756-759, doi: 10.1109/ICPEC.2013.6527756. ( <i>EI indexed and Indexed in SCOPUS</i> )                                                              |
| 35. D. Raveendhra, B. Kumar, D. Mishra and M. Mankotia, "Design of FPGA based open circuit voltage MPPT charge controller for solar PV system," 2013 International Conference on Circuits, Power and Computing Technologies (ICCPCT), Nagercoil, 2013, pp. 523-527, doi: 10.1109/ICCPCT.2013.6529012. ( <i>Indexed in SCOPUS</i> )                                                                                                         |
| 36. D. Raveendhra, P. Thakur and B. L. Narasimha Raju, "Design and small-signal analysis of solar PV fed FPGA based Closed Loop control Bi-Directional DC-DC converter," 2013 International Conference on Circuits, Power and Computing Technologies (ICCPCT), Nagercoil, 2013, pp. 283-288, doi: 10.1109/ICCPCT.2013.6529011. ( <i>Indexed in SCOPUS</i> )                                                                                |
| 37. T. Saini, D. Raveendhra and P. Thakur, "Stability analysis of FPGA based perturb and observe method MPPT charge controller for solar PV system," 2013 Students Conference on Engineering and Systems (SCES), Allahabad, 2013, pp. 1-5, doi: 10.1109/SCES.2013.6547545. ( <i>Indexed in SCOPUS</i> )                                                                                                                                    |

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| 38. | A. Chauhan, P. Thakur and D. Raveendhra, "Assessment of induction motor performance under supply voltage unbalance: A review," 2013 Students Conference on Engineering and Systems (SCES), Allahabad, 2013, pp. 1-6, doi: 10.1109/SCES.2013.6547498. ( <a href="#">Indexed in SCOPUS</a> )                                                                                                         |
| 39. | D. Raveendhra, P. Prakash and P. Saini, "Simulation based analysis of FPGA controlled Cascaded H-Bridge Multilevel inverter fed solar PV system," 2013 International Conference on Energy Efficient Technologies for Sustainability, Nagercoil, 2013, pp. 568-572, doi: 10.1109/ICEETS.2013.6533447. ( <a href="#">Indexed in SCOPUS</a> )                                                         |
| 40. | S. Singh, M. Singh, S. Chanana and D. Raveendhra, "Operation and control of a hybrid wind-diesel-battery energy system connected to micro-grid," 2013 International Conference on Control, Automation, Robotics and Embedded Systems (CARE), Jabalpur, 2013, pp. 1-6, doi: 10.1109/CARE.2013.6733758. ( <a href="#">Indexed in SCOPUS</a> )                                                        |
| 41. | D. Raveendhra, P. Joshi and R. K. Verma, "Performance and control system design for FPGA based CVMPT boost converter for remote SPV water pumping system applications," 2014 POWER AND ENERGY SYSTEMS: TOWARDS SUSTAINABLE ENERGY, Bangalore, 2014, pp. 1-6, doi: 10.1109/PESTSE.2014.6805305. ( <a href="#">Indexed in SCOPUS</a> )                                                               |
| 42. | D. Raveendhra, S. Faruqi and P. Saini, "Transformer less FPGA Controlled 2-Stage isolated grid connected PV system," 2014 POWER AND ENERGY SYSTEMS: TOWARDS SUSTAINABLE ENERGY, Bangalore, 2014, pp. 1-6, doi: 10.1109/PESTSE.2014.6805304. ( <a href="#">Indexed in SCOPUS</a> )                                                                                                                  |
| 43. | D. Raveendhra, R. Kumar and S. Singh, "Performance investigation of FPGA controlled central three-level diode clamped inverter in two-stage solar photo voltaic (SPV) system," 2014 IEEE 2nd International Conference on Electrical Energy Systems (ICEES), Chennai, 2014, pp. 206-211, doi: 10.1109/ICEES.2014.6924169. ( <a href="#">Indexed in SCOPUS</a> )                                     |
| 44. | D. Raveendhra, P. R. u. Zaman and K. Govind, "FPGA controlled high gain bi-directional DC-DC converter (BDC) for energy storage of solar power," 2014 IEEE 2nd International Conference on Electrical Energy Systems (ICEES), Chennai, 2014, pp. 300-305, doi: 10.1109/ICEES.2014.6924185. ( <a href="#">Indexed in SCOPUS</a> )                                                                   |
| 45. | D. Raveendhra and M. K. Pathak, "Modular multi-level inverter with self-healing power unbalancing capability in single stage solar PV systems," 2017 IEEE 15th Student Conference on Research and Development (SCOREd), Putrajaya, Malaysia, 2017, pp. 402-407, doi: 10.1109/SCORED.2017.8305356. ( <a href="#">Indexed in SCOPUS</a> )                                                            |
| 46. | D. Raveendhra, H. Mohan, M. K. Pathak and P. Rajana, "Model reference adaptive controller-based stand-alone solar PV pumping system," 2017 IEEE 15th Student Conference on Research and Development (SCOREd), Putrajaya, Malaysia, 2017, pp. 408-413, doi: 10.1109/SCORED.2017.8305355. ( <a href="#">Indexed in SCOPUS</a> )                                                                      |
| 47. | Dogga Raveendhra, Mohammed Mahdi, Ramy Hakim, Rached Dhaouadi, Shayok Mukhopadhyay, Nasser Qaddoumi, "Wireless Charging of an Autonomous Drone", 6th International Conference on Electric Power and Energy Conversion Systems (EPECS-2020), Istanbul, Turkey, Oct. 5 –7, 2020. ( <a href="#">Indexed in SCOPUS</a> )                                                                               |
| 48. | D. Raveendhra, Praveen. J and P. Rajana, "Capacitor Clamped Boost Inverter for Fuel Cell-based Distributed Generation system with Battery Back Up," 2021 Fourth International Conference on Electrical, Computer and Communication Technologies (ICECCT), 2021, pp. 1-6. ( <a href="#">Indexed in SCOPUS</a> )                                                                                     |
| 49. | D. Raveendhra, J. Praveen and P. Rajana, "Mitigation of Electrical Inertia of PE Converters in Solar Powered HESS system for Remote Area Power System Applications using Synergetic Controller," 2021 Fourth International Conference on Electrical, Computer and Communication Technologies (ICECCT), 2021, pp. 1-8, doi: 10.1109/ICECCT52121.2021.9616698. ( <a href="#">Indexed in SCOPUS</a> ) |
| 50. | M. Prashanth, D. Raveendhra, A. Giridhar and B. Narasimha Raju, "DC-Link Current Ripple Reduction in Switched Reluctance Machine Drives," 2022 IEEE 2nd International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2022, pp. 1-6, doi: 10.1109/SeFeT55524.2022.9909285. ( <a href="#">Indexed in SCOPUS</a> )                                    |
| 51. | M. Prashanth, D. Raveendhra, A. Giridhar and B. Narasimha Raju, "Switched Reluctance Machine Drive Analysis with Fault-Tolerant Power Converter," 2022 IEEE 2nd International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2022, pp. 1-6, doi: 10.1109/SeFeT55524.2022.9909332. ( <a href="#">Indexed in SCOPUS</a> )                            |
| 52. | S. Kalathi, D. Raveendhra and N. Raju Bl, "Efficient single-phase grid connected transformer-less inverter with active and reactive power control," 2022 IEEE 2nd International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2022, pp. 1-6, doi: 10.1109/SeFeT55524.2022.9909332. ( <a href="#">Indexed in SCOPUS</a> )                          |
| 53. | S. Kalathi, D. Raveendhra and N. Raju Bl, "Single-Phase H6 Inverter with Hybrid Modulation Scheme for Solar PV application," 2022 IEEE 2nd International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2022, pp. 1-6, doi: 10.1109/SeFeT55524.2022.9909000. 2 ( <a href="#">Indexed in SCOPUS</a> )                                               |
| 54. | D. G. Padhan, D. Raveendhra, S. Sahoo and V. Mahesh, "Comparative Analysis between Fuzzy and PR controller in Single-Phase H bridge Inverters by Power Decoupling Strategy," 2022 IEEE 2nd International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2022, pp. 1-4, doi: 10.1109/SeFeT55524.2022.9909082. ( <a href="#">Indexed in SCOPUS</a> ) |
| 55. | R. B. S. D. J, U. R. V and D. Raveendhra, "Performance and analysis of three phase SAPF under different control algorithms for power quality problems," 2022 IEEE 2nd International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2022, pp. 1-8, doi: 10.1109/SeFeT55524.2022.9909033. ( <a href="#">Indexed in SCOPUS</a> )                      |

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| 56. T. Amulya, D. Raveendhra, J. Praveen and B. Narasimha Raju, "SPV Power Conversion using Bidirectional H6 Inverter with reduced Common Mode Voltages," 2022 IEEE 2nd International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2022, pp. 1-7, doi: 10.1109/SeFeT55524.2022.9908980. ( <a href="#">Indexed in SCOPUS</a> )                             |
| 57. T. Amulya, D. Raveendhra, J. Praveen and B. Narasimha Raju, "Performance of Dual Differential Parallel Buck Grid Connected inverter for Solar Power Conversion," 2022 IEEE 2nd International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2022, pp. 1-6, doi: 10.1109/SeFeT55524.2022.9909470. ( <a href="#">Indexed in SCOPUS</a> )                  |
| 58. Shiv Prakash Singh, Suneel Raju Pendem, Dogga Raveendhra, "A Review on DC-DC Converters for Fuel Cell Electric Vehicle Applications," 2024 IEEE students Conference on Engineering and Systems (SCES-2024), June 21-23, Prayagraj, India ( <a href="#">Indexed in SCOPUS</a> )                                                                                                                          |
| 59. Abhinav Kumar Singh, Dogga Raveendhra "High Performance Solar Powered EV Charger", 2024 IEEE 4 <sup>th</sup> International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2024 ( <a href="#">Indexed in SCOPUS</a> )                                                                                                                                    |
| 60. Abhinav Kumar Singh, Dogga Raveendhra, Suneel Raju Pendem "Improved Filtering with Hybrid Damping Suppression in Grid Connected Solar PV inverter", 2024 IEEE 4 <sup>th</sup> International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2024 ( <a href="#">Indexed in SCOPUS</a> )                                                                   |
| 61. Varun Sharma, Dogga Raveendhra, Souradip Dey, "Robust Controller for EV Onboard Charger for Grid Resilience under Weak Grid Conditions" 2024 IEEE 4 <sup>th</sup> International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2024 ( <a href="#">Indexed in SCOPUS</a> )                                                                               |
| 62. Siddhant Rao, Ankita Kumari, Dogga Raveendhra, BL Narasimha Raju, Aurobinda Panda, "Efficient Tracking Control Strategy for High Voltage Gain Transformerless DC-DC Converter in DC Drive Applications", 2024 IEEE 4 <sup>th</sup> International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2024 ( <a href="#">Indexed in SCOPUS</a> )              |
| 63. Gyanendra Vaishya, Dheeraj Saini, Yugal Patidar, Dogga Raveendhra, BL Narasimha Raju, Praveen J, "Development of Characterization Circuit for Power Semiconductor Devices", 2024 IEEE 4 <sup>th</sup> International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2024 ( <a href="#">Indexed in SCOPUS</a> )                                           |
| 64. Mohammad Asif Ali, Nilesh Prakash Singh, Dogga Raveendhra, Suneel Raju Pendem, BL Narasimha Raju & Phaneendhra Babu B, "Single Stage Isolated Power Conditioning Unit for Plugin Electric Vehicles On-Board Charger", 2024 IEEE 4 <sup>th</sup> International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2024 ( <a href="#">Indexed in SCOPUS</a> ) |
| 65. Aman Keshari, Dogga Raveendhra, BL Narasimha Raju, Pradyumn Chaturvedi, U Ramanjaneya Reddy & Phaneendhra Babu B, "Switched Reluctance Motor-based EV Drive with Bidirectional Grid Interaction", 2024 IEEE 4 <sup>th</sup> International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2024 ( <a href="#">Indexed in SCOPUS</a> )                     |
| 66. Maya Vijayan, Ramanjaneya Reddy U, Tarkeshwar Mahto, Narasimharaju B L, Raveendhra Dogga, "Independently controlled single-input dual-output DC-DC converter for DC microgrid based PV-fed EV charging stations", 2024 IEEE 4 <sup>th</sup> International Conference on Sustainable Energy and Future Electric Transportation (SeFeT), Hyderabad, India, 2024 ( <a href="#">Indexed in SCOPUS</a> )     |
| 67. Mohammad Asif Ali, Aman Keshri, Dogga Raveendhra, BL Narasimha Raju, "Higher Order Sliding Robust Control Design for Single-Stage Plugin Electric Vehicle ", international Transportation Electrification Conference (ITEC), INDIA 2025, Pune, India, 18-20 September 2025. ( <a href="#">Indexed in SCOPUS</a> )                                                                                       |
| 68. Mohammad Asif Ali, Aman Keshri, Dogga Raveendhra, Sunil Raju Pendem, BL Narasimha Raju & Pradyumn Chaturvedi, "Robust Controller for Single-Stage Isolated Power Conditioning Unit On-Board Charger", The IEEE International Conference on Smart Technologies for Power, Energy, and Control (STPEC-2025), Goa, 10-13 December, 2025. ( <a href="#">Indexed in SCOPUS</a> )                             |
| 69. Aman Keshari, Mohammad Asif Ali, Dogga Raveendhra, Suneel Raju Pendem, BL Narasimharaju, "Performance investigation and characterization of High-Power Semiconductor devices using ERSC", The IEEE International Conference on Smart Technologies for Power, Energy, and Control (STPEC-2025), Goa, 10-13 December, 2025. ( <a href="#">Indexed in SCOPUS</a> )                                         |
| 70. Priya Rani, Dogga Raveendhra, "Performance Analysis and Optimization of Solar Powered Switched-Inductor Converter for High Step-Up Conversion", 2nd edition of IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCE 2026), Sohra (Cherrapunjee), Meghalaya, India, 12-14, June 2026 [Accepted].                                                                    |
| 71. Priya Rani, Dogga Raveendhra, "Performance Analysis and Optimization of Hybrid Switched-Inductor Converters for High Step-Up Conversion", 2nd edition of IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCE 2026), Sohra (Cherrapunjee), Meghalaya, India, 12-14, June 2026 [Accepted].                                                                          |
| 72. Vishruti Singh, Dogga Raveendhra, "T-Type Converter-based three-phase Universal Onboard Charger with V2X Capabilities", IEEE North-East India International Energy Conversion Conference and Exhibition(NE-IECCE 2026), Sohra, Meghalaya, India, 12-14, June 2026 (Accepted)                                                                                                                            |
| 73. Vishruti Singh, Dogga Raveendhra, "T-Type Converter-based single-phase Universal Onboard Charger with V2X Capabilities" 6th IEEE International Conference on Sustainable Energy and Future Electric Transportation (SeFet 2026), Nagpur, Maharashtra, India, 8-11, July 2026 (Accepted)                                                                                                                 |
| 74. Shoaib Akhter, Dogga Raveendhra, "Dwell Angle Estimation for Torque Ripple Reduction in Solar Powered Battery Assisted SRM Drive", 10th International Conference on Inventive Systems and Control (ICISC 2026), Coimbatore, Tamil Nadu, India, 5-7, August 2026 ( <a href="#">Accepted</a> )                                                                                                            |

75. Shoaib Akhter, Dogga Raveendhra, "Speed Control and Torque Ripple Reduction in Battery-Powered SRM Drives" 10th International Conference on Inventive Systems and Control (ICISC 2026), Coimbatore, Tamil Nadu, India, 5-7, August 2026  
(Accepted)

#### Projects

| #  | Title                                                                                                   | Agency /Scheme/period                                                 | Cost in Lakhs                                                                          | Tenure              |
|----|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------|
| 1. | Design of low cost and highly efficient testing circuit for power semiconductor device characterization | DST, Govt. of. India                                                  | 44.7                                                                                   | 2022-25             |
| 2. | Capacitor Clamped Boost Inverter for on-grid Solar PV Applications                                      | MSME, Govt. of. India                                                 | 15                                                                                     | 2022-23             |
| 3. | High Performance Switched Reluctance Motor Drive for EV applications                                    | R & C Department, MNNIT                                               | 5                                                                                      | 2023-24             |
| 4. | Innovative research in energy efficiency and hybrid energy-storage systems                              | Renewable Energy Research Center, American University of Sharjah, UAE | 38.5                                                                                   | 2019-20             |
| 5. | Capacitor Clamped Boost Inverter for Distributed Power Generation using Renewable Energy Sources        | GAIL Phank Scheme, GAIL India Ltd.                                    | 140                                                                                    | 2019-2021           |
| 6. | Capacitor Clamped Boost Inverter for Solar Power Generation                                             | TIDES Business Incubator, IIT Roorkee                                 | Prototyping grant of Rs. 80,000 + Incubation Facilities for 2 years' worth of 22 Lakhs | Aug 2017 – Aug 2020 |
| 7. | Single Stage Inverter for Solar PV System                                                               | IUSSTF, Govt. of. India                                               | 10.4                                                                                   | 2016                |
| 8. | Design of low cost and highly efficient testing circuit for power semiconductor device characterization | Measure India Corporation Private Limited                             | 3                                                                                      | 2022-23             |

#### Faculty Development Programs organised/Conducted

1. TPC Chair, SCES 2024
2. Session Chair, SCES 2024
3. Tutorial Chair, SeFeT - 2022
4. **Co-Ordinator** for Online Faculty Development Programme (FDP) on "Modelling, Simulation and Control of Advanced Power Converters" during 28th March – 06th April, 2022 Organized by E & ICT Academy & Department of Electrical Engineering, NIT Warangal in association with Gokaraju Rangaraju Institute of Engineering and Technology, Hyderabad (Sponsored by Ministry of Electronics and Information Technology (MeitY), GOI).
5. **Co-Ordinator** for One Week Faculty Development Programme on "Recent Developments in Power Electronics and its Applications", organized by Electrical and Electronics Engineering Department, Vardhaman College of Engineering, Hyderabad, during 31<sup>st</sup> Aug 2020 to 5th Sep 2020, in Association with Zunik Energies Pvt. Ltd.
6. **Convener** for Multi-Agent Systems and Distributed Control (MAS 2024) September 27-31, 2024 Organized by Department of Electrical Engineering Motilal Nehru National Institute of Technology
7. **Co-Ordinator** for Online Faculty Development Programme (FDP) on "Power Electronics, Energy Storage and Renewable Technologies (PEESRT) for E-Transportation in India (Basic)
8. **Co-Ordinator** for Online Faculty Development Programme (FDP) on "Power Electronics, Energy Storage and Renewable Technologies (PEESRT) for E-Transportation in India (Advanced)
9. **Co-Ordinator** for One Week Faculty Development Programme on "Recent Developments in Power Electronics", organized by Electrical and Electronics Engineering Department, Aditya Engineering College, Surampalem, during 12<sup>th</sup> Nov 2018 to 17<sup>th</sup> Nov 2018, in Association with Zunik Energies Pvt. Ltd.
10. **Co-Coordinator** for AICTE Sponsored Two Week FDP (Online-Phase-I) on "Sustainable Technologies for Electric Transportation Systems" scheduled on 14<sup>th</sup> June 2021 – 26<sup>th</sup> June 2021.

#### Awards/ Honoured Positions

| #  | Award/Honoured Position | Awarded/ Honoured By                                                                                                                                                                                                                              | Year          |
|----|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1. | Resource Person         | One Week Short Term Training Programme, "AI/ML Application for Renewable Energy Systems", National Institute of Technology, Delhi (Hybrid Mode), <b>01st – 05th June 2026</b>                                                                     | June 2026     |
| 2. | Expert lecture          | "Emerging Trends in Power Electronics and Electric Vehicle Mobility." at United College of Engineering and Research, Prayagraj, 11 April 2026                                                                                                     | April 2026    |
| 3. | Resource Person         | Conducting One-Week Online Faculty Development Program in Collaboration with CITD – MSME, "Smart EV Control: AI, IoT and V2X Technologies, GMR Institute of Technology (Deemed to be University), 6 <sup>th</sup> to 10 <sup>th</sup> April 2026. | April 2026    |
| 4. | Resource Person         | DSP Controlled Power Supply Design for Drones, One-Week Workshop on IoT and Embedded Systems for Smart Drones (IESSD-2026), 23 <sup>rd</sup> – 27 <sup>th</sup> February 2026                                                                     | February 2026 |

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|-----|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 5.  | Resource Person                                              | One-Week Short Term Course on “Resilient and Intelligent Systems for Empowering Viksit Bharat 2047 (RISE- 2026)”, 16th – 20th February 2026                                                                                | February 2026 |
| 6.  | Resource Person                                              | “A Full-Stack Approach to Power Electronic Hardware Development”, BVRIT HYDERABAD College of Engineering for Women                                                                                                         | January 2026  |
| 7.  | Resource Person                                              | "Modelling and Simulation of Renewable Energy Systems" at Rajiv Gandhi University of Knowledge Technologies, Srikakulam on 24/12/2025                                                                                      | Dec 2025      |
| 8.  | Resource Person                                              | 5th INTERNATIONAL CONFERENCE ON RECENT ADVANCES IN MATHEMATICAL SCIENCES AND APPLICATIONS (RAMSA-25) at Gayatri Vidya Parishad College of Engineering (Autonomous). 19th to 22nd DECEMBER 2025                             | Dec 2025      |
| 9.  | Member Board of Studies (BoS)                                | KL University                                                                                                                                                                                                              | Dec 2025      |
| 10. | Resource Person                                              | AICTE–ATAL Faculty Development Programme (FDP) on Advanced Technologies and AI/ML Integration in Microgrids and Electric Vehicles at Guru Nanak Institute of Technology (GNIT)                                             | Dec 2025      |
| 11. | Resource person                                              | FDP on “Intelligent Power Electronics and Cyber-Physical Energy Systems for a Sustainable Future” at Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology                                            | Nov 2025      |
| 12. | Resource person                                              | Online Faculty Development Programme (FDP) on Advancements in Electric Vehicles and Autonomous Mobility for a Sustainable Future (AEVAMSF) at BVRIT Hyderabad and NIT Warangal                                             | July 2025     |
| 13. | Speaker                                                      | AICTE ATAL Online 6-Day Faculty Development Programme on “Emerging Trends and Future Prospects in Advanced Energy Conversion Technologies and Applications at VIT Bhimavaram                                               | Aug 2025      |
| 14. | Expert Talk                                                  | Online FDP on "Power Electronics Technology for Self-Reliant and Viksit Bharat at NIT Warangal                                                                                                                             | Feb 2025      |
| 15. | Resource person                                              | FDP on “Leveraging Advanced Power Conversion Systems and Artificial Intelligence for Electric Vehicles at NIT Patna                                                                                                        | May 2025      |
| 16. | Resource person                                              | FDP on Design of Advanced Controller for Microgrid and Electric Vehicles at Vignan Lara Institute of Technology                                                                                                            | Jan 2025      |
| 17. | Expert Lecture                                               | the MSME-sponsored Five-Day Advanced-ESDP Programme on “Renewable Energy Sources in Electric Vehicle and Distributed Generation Systems at NIT Sikkim                                                                      | Feb 2025      |
| 18. | Member Board of Studies (BoS)                                | KL University                                                                                                                                                                                                              | Aug 2025      |
| 19. | BOS Member                                                   | EEE Department, KLU                                                                                                                                                                                                        | 2024          |
| 20. | Resource Person                                              | Sustainable Development: Intersection of Green Energy and eMobility (IGEEM-2024)” at NIT Rourkela                                                                                                                          | 2024          |
| 21. | DCC Sub-Committee Member                                     | Quantum Technology Division, Telecommunication Engineering Center                                                                                                                                                          | 2024          |
| 22. | Resource Person                                              | “AI&ML Tools for Material Science & Electrical Engineering Applications”, Gayatri Vidya Parishad College of Engineering, Vizag                                                                                             | 2024          |
| 23. | National Webinar as a Resource Person                        | Webinar on “Electrifying Experiences: Design-Thinking the Future of EV Mobility”, K.S. Rangasamy College of Technology, Tamilnadu                                                                                          | 2024          |
| 24. | BOS Member                                                   | EEE Department, KLU                                                                                                                                                                                                        | 2024          |
| 25. | Resource Person                                              | Anil Neerukonda Institute of Technology and Sciences, Andhra Pradesh.                                                                                                                                                      | 2023          |
| 26. | Resource Person                                              | Delivered an Invited Lecture in ATAL sponsored FDP entitled on Certain Aspects of Electric Vehicles to achieve Sustainable Energy, Vignan Lara Institute of Technology, Vadlamudi                                          | 2023          |
| 27. | Keynote Speaker                                              | DST-SERB Sponsored Conference, KL University                                                                                                                                                                               | 2023          |
| 28. | Resource Person                                              | Delivered an Invited Lecture in ATAL sponsored FDP entitled “AI & Smart Technology Applications to Solar energy and Electrical vehicles Technology - Challenges, Issues and Hands-on" (AIST-SET), ATAL-FDP, BVRIT Narsapur | 2023          |
| 29. | <b>BOS Member</b>                                            | EEE Department, KLU                                                                                                                                                                                                        | 2023          |
| 30. | Seminar                                                      | Machine Learning, Computer Vision, and Dynamic Systems Group, The Skolkovo Institute of Science and Technology, Russia                                                                                                     | 2023          |
| 31. | Invited Session                                              | FDP, MNNIT                                                                                                                                                                                                                 | 2023          |
| 32. | Invited Session                                              | FDP, GRIET                                                                                                                                                                                                                 | 2023          |
| 33. | Invited Session                                              | FDP, GRIET                                                                                                                                                                                                                 | 2023          |
| 34. | Invited Session                                              | FDP, MNIT Jaipur                                                                                                                                                                                                           | 2022          |
| 35. | Invited Session                                              | FDP, SVNIT                                                                                                                                                                                                                 | 2022          |
| 36. | Invited Session                                              | FDP, SVNIT                                                                                                                                                                                                                 | 2022          |
| 37. | Excellence in Doctoral Research Award (for best Ph. D. Work) | IIT Roorkee                                                                                                                                                                                                                | 2020          |
| 38. | BOS Member                                                   | VNIT Nagpur                                                                                                                                                                                                                | 2022          |

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|-----|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 39. | Resource person in a Faculty Development programme                              | Eight-day online FDP on "Modelling, Simulation & Control of Advanced Power Converters", at GRIET                                                                                      | 2022 |
| 40. | Resource person in a Faculty Development programme                              | Eight-day online FDP on "Power Electronics for Electric Vehicles and Renewable Energy Systems during <b>16<sup>th</sup> – 24<sup>th</sup> March, 2022</b>                             | 2022 |
| 41. | Resource person in a Faculty Development programme                              | Eight-day online FDP on "Applications of Power Electronics in Electric Vehicles & Energy Storage", at NITK                                                                            | 2022 |
| 42. | Resource person in AICTE-ISTE Sponsored Induction/Refresher Induction Programme | Online FDP on "Recent Trends in Electric Vehicles" 17 <sup>th</sup> – 23 <sup>rd</sup> February 2022 in Online Mode at Vardhaman College of Engineering, Hyderabad, Telangana, India. | 2022 |
| 43. | Resource person in AICTE-ISTE Sponsored Induction/Refresher Programme           | AICTE/ ISTE Refresher Programme (ONLINE) - Micro Grid: Integration and Power Quality Issues - Dec. 7-13, 2021                                                                         | 2021 |
| 44. | Resource person in a Faculty Development programme                              | CEP, NIT Warangal Sponsored Five-day online FDP on Potential Research Trends in Advanced Energy Conversion Technologies and Applications, December 06 - 10, 2021 at VIT Bhimavaram    | 2021 |
| 45. | Resource person in a Faculty Development programme                              | ATAL Academy Sponsored Online Faculty Development Programme (FDP) on "Electric Vehicle (EV) : Challenges and Technology Advancements" by NSIC, Hyderabad                              | 2021 |
| 46. | Resource person in a Faculty Development programme                              | AICTE sponsored ATAL FDP on "ELECTRIC TRANSPORTATION INFRASTRUCTURE FOR E-MOBILITY IN INDIA" to be held during 02 to 06, Aug 2021                                                     | 2021 |
| 47. | Resource person in a Faculty Development programme                              | AICTE Sponsored Two Week FDP (Online-Phase-I) on "Sustainable Technologies for Electric Transportation Systems, GRIET                                                                 | 2021 |
| 48. | Guest Lecture                                                                   | Guest Lecture - Start-up Cell - KSRMCE                                                                                                                                                | 2021 |
| 49. | Guest Speaker                                                                   | AICTE –ATAL Sponsored FDP on "ELECTRIC VEHICLES- RESEARCH ISSUES AND CHALLENGES" GRG POLYTECHNIC COLLEGE                                                                              | 2021 |
| 50. | BOS Member                                                                      | KL University                                                                                                                                                                         | 2021 |
| 51. | Keynote Speaker in FDP                                                          | Government Engineering College Raipur, Chhattisgarh                                                                                                                                   | 2020 |
| 52. | Keynote Talk                                                                    | Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal                                                                                                                                     | 2020 |
| 53. | Keynote Speaker                                                                 | Government Engineering College Raipur, Chhattisgarh                                                                                                                                   | 2020 |
| 54. | Resource person in a Faculty Development programme                              | ATAL FDP/NMREC                                                                                                                                                                        | 2020 |
| 55. | Resource person in a Faculty Development programme                              | ATAL FDP at Kallam Harinadh Reddy Institute of Technology                                                                                                                             | 2020 |
| 56. | Resource person in a Faculty Development programme                              | Vardhaman Engineering College Hyderabad                                                                                                                                               | 2020 |
| 57. | Guest Lecture                                                                   | KL University, Viziawada                                                                                                                                                              | 2020 |
| 58. | Guest Lecture                                                                   | Vishakha Institute of Engineering & Technology                                                                                                                                        | 2020 |
| 59. | Resource person in a Faculty Development programme                              | ATAL FDP at MVGR College of Engineering                                                                                                                                               | 2020 |
| 60. | Best Innovation Award                                                           | Zunik Energies Pvt. Ltd                                                                                                                                                               | 2019 |
| 61. | Who's Who in the World 2019                                                     | Marquis, USA                                                                                                                                                                          | 2019 |
| 62. | Resource person in a Faculty Development programme                              | NIT, Warangal                                                                                                                                                                         | 2019 |
| 63. | Industry Expert for Board of Studies                                            | NIT Nagpur                                                                                                                                                                            | 2019 |
| 64. | Albert Nelson Marquis Lifetime Achievement Award                                | Marquis Who's Who (USA)                                                                                                                                                               | 2018 |
| 65. | Who's Who in the World 2018                                                     | Marquis, USA                                                                                                                                                                          | 2018 |
| 66. | Resource person in a Faculty Development programme                              | Aditya Group of Engineering Colleges                                                                                                                                                  | 2018 |
| 67. | Guest Lecture                                                                   | Dev Bhoomi Group of Institutes                                                                                                                                                        | 2018 |
| 68. | Expert Lecture                                                                  | NITTTR, Chandigarh                                                                                                                                                                    | 2018 |
| 69. | Delegate for India-Italy Summit                                                 | DST-CII, New Delhi                                                                                                                                                                    | 2018 |
| 70. | Most Promising Innovation Award (1 <sup>st</sup> Prize)                         | IKMC 2018 Start-up Exhibition                                                                                                                                                         | 2018 |
| 71. | Mentor for start-ups                                                            | SAIC, Thapar University                                                                                                                                                               | 2018 |
| 72. | Session Chair                                                                   | IEEE SCORed International Conference, Malaysia                                                                                                                                        | 2017 |

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|-----|----------------------------------------------------|----------------------------------------|---------|
| 73. | International Travel Grant                         | IIT Roorkee                            | 2017    |
| 74. | Resource person in a Faculty Development programme | MVGR College of Engineering            | 2017    |
| 75. | Judges Choice award                                | TIDES-IITRHF                           | 2017    |
| 76. | Bhaskara Advanced Solar Energy Fellowship          | DST, India                             | 2016    |
| 77. | MHRD scholarship for M. Tech                       | Ministry of Human Resource Development | 2010-12 |
| 78. | MHRD scholarship for Ph.D.                         | Ministry of Human Resource Development | 2014-19 |

#### Reviewer for Journals

1. [IEEE Transactions on Industrial Electronics](#) (Impact factor of 6.498)
1. [IEEE Transactions on Journal of Emerging and Selected Topics in Power Electronics](#) (Impact factor of 5.177)
2. [International Journal of Electrical Power and Energy Systems](#), Elsevier (Impact factor of 3.432)
3. [Journal of Solar Energy](#), Elsevier (Impact factor of 3.469)
4. [Power Electronics](#), IET (Impact factor of 1.6)
5. [Generation, Transmission & Distribution](#), IET (Impact factor of 1.353)
6. [International Transactions on Electrical Energy Systems](#) (Impact factor of 1.085)
7. [Renewable Power Generation](#), IET (Impact factor of 1.60)
8. [Electric Power Components and Systems](#) Journal, Taylor and Francis (Impact factor of 0.69)
9. Reviewer for [IETE Journal of Research](#) (Impact factor of 0.6)
10. Reviewer for [IEEE Access](#)
11. Reviewer for [Mathematics and Computers in Simulation](#) (Elsevier)
12. [Bulletin of Electrical Engineering and Informatics](#) (BEEI)
13. [Journal of TELKOMNIKA](#) (Telecommunication, Computing, Electronics and Control)
14. Indonesian Journal of [Electrical Engineering and Computer Science](#)
15. International Journal of [Electrical and Computer Engineering](#) (IJECE)
16. International [Journal of Advances in Applied Sciences](#) (IAAS)
17. [Journal of Computing and Digital Systems](#)
18. [Journal of Trends in Computer Science and Information Technology](#)
19. [International Journal of Robotics and Automation](#) (IAES)
20. [International Journal of Computing and Digital Systems](#) (IJCDs)
21. [IETE Journal of Research](#)
22. [International Journal of Green Energy](#)
23. [International Journal of Emerging Electric Power Systems](#) (IJEPS)
24. [International Journal of Electrical Power and Energy Systems](#)
25. [International Journal of Engineering Science and Technology](#)

#### Reviewer for IEEE Conferences

1. Reviewer for [IEEE International Conference on Research & Development](#) (SCORED), 16-17 December 2013, Putrajaya, Malaysia.
2. Reviewer for IEEE [International Conference on Composite materials and Renewable Energy Applications](#) ICCMERA'2014, 22-24 January 2014, Sousse, Tunisia.
3. Reviewer for 2014 IEEE 8th [International Conference on International Power Engineering and Optimization](#) (PEOCO2014), 24-25 March 2014, Langkawi, Malaysia.
4. Reviewer for 23rd IEEE [International Symposium on Industrial Electronics](#) (ISIE 2014), June 1-4, 2014, Istanbul, Turkey.
5. Reviewer for 3rd International [Conference on Frontiers in Intelligent Computing, Theory and Application](#) (FICTA-2014), Orissa, India, 14-15 November 2014.
6. Reviewer for [Annual Convention and International Conference on Emerging ICT for bridging future](#) (CSI-2014), 12 Dec 2014.
7. Reviewer for [Second International Conference on Information systems Design and Intelligent Applications](#) - 2015 (INDIA-2015) Kalyani, India, January 8-9, 2015
8. Reviewer for [Renewable Energy and Green Technology International Conference](#) 2015 (REEGETECH'2015), Bali, Indonesia, 2 – 4 June 2015.
9. Reviewer for 2015 [Advanced Research in Material Sciences, Manufacturing, Mechanical and Mechatronic Engineering Technology International Conference](#) (AR4MET2015), Bali, Indonesia, 2-4 June 2015
10. Reviewer for 2017 [IEEE SCORED](#) held during 13th Dec 2017 to 14th Dec 2017 at Everly Hotel, Putrajaya, Malaysia
11. Reviewer for [CENCON 2017](#)-(2017 IEEE Conference on Energy Conversion (CENCON)) Kuala Lumpur, Malaysia on 30–31 October 2017
12. Reviewer for [ARiET'2017] 2017 2nd [Advanced Research in Electrical and Electronic Engineering Technology](#) (ARiET)

13. Reviewer for [RESEEC'S'2017] 2017 [Recent Development in Sciences, Engineering, and Computer Sciences International Conference](#)
14. Reviewer for [I4CT'2018] 2018 4th [International Conference on Computer, Communication and Control Technology](#)
15. Reviewer for [SISTECH'2018] 2018 [Symposium on Islamic Sciences and Technology](#)
16. Reviewer for [AVAREIT'2018] 2018 2nd [Advanced Research in Electronic Engineering and Information Technology International Conference](#)
17. Reviewer for the 2018 [International Conference on Advances in Computing, Communications, and Informatics](#) (ICACCI)
18. Reviewer for ICCMREA 2017 (2017 [International Conference on Composite Materials & Renewable Energy Applications](#))
19. Reviewer for [Green Energy Conference](#) 2018
20. Reviewer for [Conference on Engineering Education](#) (ICEED 2018)
21. Reviewer for [IEEE International Conference on Advanced Computational and Communication Paradigms](#) (ICACCP-2018)
22. Reviewer for [IEEE PES Asia-Pacific Power and Energy Engineering Conference](#) (APPEEC 2018)
23. Reviewer for [International Conference on Computing and Network Communications](#) (CoCoNet'19)
24. Reviewer for [IEEE Conference on Energy Conversion](#) (CENCON 2019)
25. Reviewer for International [Conference on Advanced Computational and Communication Paradigms](#) (ICACCP 2019)
26. Reviewer for [IEEE Transportation Electrification Conference](#) (ITEC 2019)
27. Reviewer for [International Symposium on Computer Vision and Machine Intelligence in Medical Image Analysis](#) (ISCMM-2019)
28. Reviewer for [IEEE International Conference on Power and Energy](#) (PECon 2020)
29. Reviewer for [Conference on Internet of Things and Embedded Intelligence](#), 2020
30. Reviewer for [IEEE Toronto International Conference on Power and Energy](#) (TICPE 2020)
31. Reviewer for [International Conference on Electrical Engineering, Computer Science and Informatics](#) (EECSI 2020)
32. Reviewer for [Ahmad Dahlan International Conference Series on Engineering, Science and Information Technology](#) 2020
33. Reviewer for [Conference on Internet of Things and Embedded Intelligence](#) 2020
34. Reviewer for [IEEE International Conference on Power and Energy](#) (PECON 2020)

#### Invited Talks

1. [Speaker Invitation](#) for INAIT Conference-2019, University of Cambridge, United Kingdom.
2. [Speech Invitation](#) from 7th Annual World Congress of Advanced Materials (W.C.A.M) 2018, in Xiamen, China with the theme of "Innovation, Integration, Transformation, and Sustainability"
3. Got an Opportunity to present an [Invited paper](#) at MIC-Electrical 2014 (1<sup>st</sup> International Conference on Electrical Engineering and Application), to be held in Athens, Greece: 4-6 April 2014.
4. Got an [invitation to deliver](#) a speech at 1st International Symposium on Energy Challenges and Mechanics, Aberdeen, Scotland, UK, 8-10 July 2014.
5. Got an [invitation to contribute a talk in a Session](#) 02: Renewable energy at 3rd International Symposium on Energy Challenges and Mechanics (ECM3) - towards a big picture, 7-9 July 2015, Aberdeen, Scotland, UK
6. [Speech Invitation](#) from the World Congress of Smart Energy-2017, Wuxi, China for Session 301: Solar PV Technologies
7. [Speaker Invitation](#) for 6th Annual World Congress of Advanced Materials-2017 (WCAM-2017, Xi'an) at Track702: Photovoltaics, Solar Energy, Artificial Photosynthesis Materials and Devices.

#### Editor/ Associate Editor/ Lead Guest Editor

1. [Associate Editor](#) for International Journal of Applied Power Engineering (IJAPE), ISSN 2088-8708, e-ISSN 2722-2578, p-ISSN 2252-8792, e-ISSN 2722-2624 (open access and free of charge journal).
2. Got an Invitation from [SciencePG](#) to propose a Special Issue in my research field and offered [Lead Guest Editor](#), Science Publishing Group, NEW YORK, U.S.A.

#### Technical Program Committee for Journals, Government Summits & Conferences

1. Technical Program Committee for IEEE [International Conference on Smart Technologies for Power, Energy and Control](#) (STPEC), held in the Department of Electrical Engineering, Visvesvaraya National Institute of Technology, Nagpur, M.S., India during 25th to 26th September 2020
2. Technical Program Committee for IEEE [International Conferences on Composite materials and Renewable Energy Applications](#) (ICCMERA'2015), Sousse, Tunisia, 22-24 January 2015.
3. Technical Program Committee for [International Conference on Solar Energy and Building](#) (ICSoEB' 2015).
4. Program Committee for 2014 IEEE International [Conference on Intelligent Energy and Power Systems](#), June 2-6, 2014, Kyiv, Ukraine.
5. Technical Program Committee for IEEE [International Conferences on Composite materials and Renewable Energy Applications](#) (ICCMERA'2014), Sousse, Tunisia, 22-24 January 2014
6. Technical Program Committee for [International Conference on Computer Vision and Image Analysis applications](#) (ICCVIA' 2014), 25-27 March 2014, Ras Al Khaimah, UAE.

7. Technical Program Committee for 2014 [IEEE Conference on Energy Conversion](#) (CENCON 2014), 13–15 October 2014, Johor Bahru, Malaysia
8. Technical Program Committee member for [IEEE International Conference on Energy Conversion](#) (CENCON 2015), Johor Bahru Malaysia, 19-21 October 2015.
9. Technical Program Committee member for 2017 [IEEE SCORed](#) to be held during 13th Dec 2017 to 14th Dec 2017 at Everly Hotel, Putrajaya, Malaysia
10. Technical Program Committee member for [CENCON 2017](#)-(2017 IEEE Conference on Energy Conversion (CENCON)) Kuala Lumpur, Malaysia on 30–31 October 2017
11. Member for [SIRS-2017](#) (Third International Symposium on Signal Processing and Intelligent Recognition Systems (SIRS'17))
12. Member for [IEEE TENSYP](#) 2017 (2017 IEEE Region 10 Symposium (TENSYP))
13. Technical Program Committee for [International Biometrics & Smart Government Summit](#) (IBMSGs' 2014), 22-24 March 2014, Dubai, UAE.
14. Technical Program committee member for the [International Journal of Computing and Digital Systems \(IJCDs\)](#).

#### Professional Memberships

1. [IEEE](#) Member
2. Member, Subcommittee of "Resonant and Soft Switching Converters" of IES Power Electronics Technical Committee (IEEE IES PETC)
3. Full Member ([MEEC](#)) of the European Energy Centre

#### Technical Qualifications

|                                  |                                                                                                                                                                                       |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>                 | C, MATLAB, VHDL                                                                                                                                                                       |
| <b>Software Packages</b>         | Xilinx System Generator, Altium, Eagle, MULTISIM, MATLAB, LabVIEW, PSIM, dSPACE Control desk, MS Office, Arduino IDE, and CCStudio                                                    |
| <b>Operating systems</b>         | Windows, Ubuntu                                                                                                                                                                       |
| <b>Hardware Controllers Used</b> | FPGA Controller (Spartan-3E and Spartan-3A), DSP Controller (TMS320F2812), dSPACE, Analog Controllers, Arduino MCUs (Arduino Due, Arduino Mega, Arduino Uno) and TI Piccolo lunch Pad |
| <b>Hardware's designed</b>       | Capacitor Clamped Boost Inverter, Multi-Level Inverter, DC-DC Converter                                                                                                               |

#### Research Works

1. **LC Impedance Source Bi-Directional Converter (Renewable Energy Centre, American University of Sharjah, UAE)**  
This project proposes an LC (Inductor and Capacitor) impedance source bi-directional DC–DC converter by redesigning after rearranging the reduced number of components of a switched boost bi-directional DC–DC converter. This new converter with a conventional modulation scheme offers several unique features, such as a) a lower number of components and b) reduced voltage stress on the capacitor compared to existing topologies. The reduction of capacitor voltage stress has the potential of improving the reliability and enhancing converter lifespan. An analysis of the proposed converter was completed with the help of a mathematical model and state-space averaging models. The converter performance under different test conditions is compared with the conventional bi-directional DC–DC converter, Z-source converter, discontinuous current quasi-Z-source converter, continuous current quasi-Z-source converter, improved Z-source converter, switched boost converter, current-fed switched boost converter, and quasi switched boost converter in the Matlab Simulink environment. MATLAB/Simulink results demonstrate that the proposed converter has lesser components count and reduced capacitors' voltage stresses when compared to the topologies mentioned above. A 24 V to 18 V LC-impedance source bi-directional converter and a conventional bidirectional converter are built to investigate the feasibility and benefits of the proposed topology. Experimental results reveal that capacitor voltage stresses, in the case of proposed topology are reduced by 75.00% and 35.80% in both boost and buck modes, respectively, compared to the conventional converter circuit.
2. **Super capacitor Characterization (Renewable Energy Centre, American University of Sharjah, UAE)**  
This project presents a simplified supercapacitor model and a universal adaptive stabilization, optimization (UAS+O) based parameter identification technique. Analytic solutions for the description of supercapacitors current, voltage, subject to cyclic voltage and current sources of varying amplitudes and frequency, consistent with electric vehicle driving cycles, are developed. Supercapacitor I-V relationships show hysteresis, indicating simultaneous energy storage and dissipation mechanisms. A reduced equivalent circuit model is proposed to accurately represent hysteresis I-V characteristics. The proposed UAS+O based technique for estimating model parameters, is supported by mathematical proofs, simulation, and experimental results.
3. **Novel Testing Circuit for Power Electronic Device Characterization: (FREEDM Systems Centre, NC State University)**  
A novel energy recirculation circuit utilizes the concept of energy recirculation, with the power augmentation capability, is proposed. Proposed ERSC can be used as a device in-situ testing unit, by utilizing naturally occurring high electrical stresses on devices from a low-cost low-power source to supply energy for high power testing of power electronic devices. This topology allows devices to be examined at full-power stresses without connected with high power load and also without demanding high

power, by storing and recirculating the energy of the energy storage elements, which elevates the capability of source power. This converter can operate in four different modes of operation, namely, soft start, magnetize, charge and energy recirculation modes with the four active states of operations attained by two active switches of the proposed converter. Another feature of this converter is, based on the devices under test devices performance these two circuits can be designed to operate in a synchronous or asynchronous manner, i.e., faster devices or slower devices can be tested.

For in-situ testing of high current and/or voltage devices, this circuit offers several advantages such as simple circuit design, does not demand any high voltage step-up transformer for realizing high voltages/currents, can be able to operate from low voltage/power supply, cheapest solution and also provides fast response in comparison with conventional cascaded boost/buck and cascaded buck/boost ERSCs. To validate the concept, simulations are carried for testing of 1.2KV and 100A using 100V and 8A. And, the same is going to prove with the help of a lab-made prototype.

#### **4. Development of Efficient Single-Stage Power Converter System for Solar PV applications (IITR)**

1. Design and modelling of single-phase and three-phase capacitor clamped boost dc-ac inverters, which can develop a power conditioning system without demanding series-connected PV modules due to its intrinsic boost capability.
2. To address the issue related to safety requirements in a solar PV system without demanding more power electronic switches to provide a dual grounding concept, a high gain power conditioning system can be developed.
3. Efficiency is increased and overall costs are lowered by achieving MV grid integration with only one step-up transformer, the inverter includes an MV-level transformer with a high turn's ratio. Large-scale PV system architectures with reduced transformer stages can be realized by using high-gain and high-quality power conditioning systems [26]. To meet this objective three phase's high gain and high-quality power conditioning system has been proposed.
4. To develop a solution for the power decoupling issue and to increase the conversion efficiency by reducing the ripples at the input side, a waveform control method can be adopted in the proposed converter. In this method, the low-frequency power pulsation caused by ac output is absorbed by the capacitors while the input source (solar) kept a constant supply of dc power to the capacitors, thereby eliminating the effect of low-frequency ripple from affecting the properties of source (solar). Since capacitor voltages can be large without affecting the desired ac output voltage, capacitors of the inverter can be minimized. This allows the use of film capacitors over electrolytic capacitors, thereby improving the inverter's lifetime.
5. Modelling of the proposed converter along with sliding mode controller for dynamic control, Reference tracking, Disturbance rejection and Wide range of control. To achieve high-performance feedback control for the system, an accurate and well-defined model is required. The small-signal model of the proposed converter will be developed so that the sliding mode controller design can be done by using the non-linear control theory. Since SMC is normally used to control boost inverter and boost inverter also suffers from the above mention problems. Solving this issue will have a significant positive impact on sliding mode controller-based boost inverters or buck-boost inverters. To address this problem Fixed frequency sliding mode controller is proposed.
6. System development and Experimentation: Prototype hardware will be developed for the proposed converter using a digital controller platform (FPGA/dSPACE kit) for control of power devices. Some of the simulation's results of the proposed converter system will be validated through this hardware set up.

#### **5. Improved Power Quality Transformerless Power Converters for Solar PV applications (IITR)**

To further enhance the conversion gain, coupled inductor-based capacitor clamped boost inverter has been proposed. This chapter introduces coupled inductor-based capacitor clamped 3-phase DC-AC boost inverter with an intrinsic step-up ability by utilizing the small passive components. The main objective of this proposed inverter is to reduce the usage of capacitors (generally preferred to increase the gain), which are a weakest reliable element in the inverter design. By shifting the filter components from ac side to intermediate place, inverter can attain boost capability as well as good power decoupling ability, since shifted capacitor acting as a good power decoupling element. Furthermore, modulating waveforms are altered such a way that voltages across the capacitors are decreased. Reduced voltage stresses on capacitors lead to better reliability and enhanced lifespan of the inverter. This inverter performance under different test conditions is compared with boost converter fed voltage source inverter (BVSI), Z-Source Inverter (ZSI), quasi Z-Source Inverter (q-ZSI), Switched Boost Inverter (SBI), Current-Fed SBI (CF-SBI), quasi SBI (qSBI), Improved ZSI (IZSI), Switched Inductor-ZSI (SL-ZSI), Switched Inductor-qZSI (SL-qZSI), Diode Assisted qZSI (DA-qZSI), Capacitor Assisted qZSI (CA-qZSI), Extended Boost ZSI (EB-ZSI), Extended Boost qZSI (EB-qZSI) and CCBI; MATLAB Simulink results demonstrated that proposed inverter capabilities are superior to above-mentioned topologies. A 1200W experimental prototype has been built to validate the feasibility and benefits of the system. Simulation and experimental results reveal that proposed inverter offers better power quality, reliability and high lifetime.

#### **6. Development of Converter System for Solar PV Power Generation: (IITR)**

Renewable energy resources will be an increasingly important part of power generation in the new millennium. The main objective of this project is to develop a power conditioning system, which can be used to extract the variable DC power from the sun. It is converted into fixed dc by using DC-DC boost converter and then converted into AC power by using 3- level diode clamped inverter, to feed AC load effectively. In this project simulation results of the FPGA Controlled Photovoltaic (PV) power conditioning system for AC, loads are presented. The power conditioning system consists of a diode clamped three-level inverter fed by a closed-loop voltage-controlled DC-DC Boost converter. This closed-loop control of the DC-DC converter is implemented by a

conventional Pulse Width Modulator (PWM) with a duty cycle ratio control method. This boost converter is designed to obtain regulated voltage from the variable DC supply. A level shifts sinusoidal PWM is used to control the multilevel inverter on FPGA.

#### 7. Simulation of SPMC based Cycloconverter (6months)

The backbone of the modern industry is electrical drives. Tremendous changes in speed control of electrical machines have been taken place with the development of power electronics. Nowadays, AC motors are also used for variable speed applications. At present in speed control of AC machines, single phase Cyclo Converters are widely used for AC#AC power conversion. In this project, Single Phase Matrix Converter (SPMC) Topology using IGBT#s as power switches is used for Cyclo Converters operation in place of the conventional thyristor as it offers many advantages. In this project, a well-known Sinusoidal Pulse Width Modulation (SPWM) scheme is used for generating gate pulses to IGBT#s. This project presents work on modelling and simulation of Cyclo Converters using MATLAB/Simulink incorporating the Sim Power System Block set.

#### Industrial Training

Attended an Industrial Training for 1 month on 'Protection and Detection of 3-Phase Transformers faults and its diagnosis' at WICORE PVT Industries Ltd. in Dec, 2008.

#### Guided Dissertations (12):

Guided 8 M. Tech Projects and 25 students for their Academic B. Tech Projects

| #   | Duration          | Name of the Candidate              | Name of the dissertation                                                                                                              | University                       |
|-----|-------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 1.  | July'12 – June'13 | Mr. Prachetha Prakash              | Development of Cascaded H-Bridge MLI for solar PV systems                                                                             | Graphic Era University, Dehradun |
| 2.  | July'12 – June'13 | Ms. Minakshi Rawat                 | Development of Bi-Directional DC-DC Converter for Solar Applications                                                                  | Graphic Era University, Dehradun |
| 3.  | July'12 – June'13 | Ms. Tripti Saini                   | Development of MPPT charge controller for photovoltaic solar PV system                                                                | Graphic Era University, Dehradun |
| 4.  | July'13 – June'14 | – Mr. Prawal Joshi                 | Performance and Control System Design for FPGA based CVMPT Boost Converter for Remote SPV Water Pumping System Applications           | Graphic Era University, Dehradun |
| 5.  | July'13 – June'14 | Mr.Parwez Ridwan                   | FPGA Controlled High Gain Bi-Directional DC-DC Converter (BDC) for Energy Storage of Solar Power                                      | Graphic Era University, Dehradun |
| 6.  | July'13 – June'14 | – Mr. Rahul Kumar                  | Performance Investigation of FPGA Controlled Central Three-Level Diode Clamped Inverter in Two-Stage Solar Photo Voltaic (SPV) System | Graphic Era University, Dehradun |
| 7.  | July'13 – June'14 | – Mr. Saad Faruqui (Co-Supervisor) | Transformer Less FPGA Controlled 2-Stage Isolated Grid-Connected PV System                                                            | Graphic Era University, Dehradun |
| 8.  | July'17 – June'18 | – Mr. Surya Kiran (Co-Supervisor)  | Common Mode Voltage Reduction Techniques in Induction Motor Drive                                                                     | JNTU Kakinada                    |
| 9.  | July 2021         | V Ranadeep                         | Control of Capacitive Coupled Impedance source type Wireless Power Transfer Network                                                   | GRIET, Hyderabad                 |
| 10. | July 2024         | Varun Sharma                       | Bidirectional AC-DC Converters for EV Chargers with PFC and V2G Capabilities                                                          | MNNIT, Allahabad                 |
| 11. | July 2024         | Abhinav Kumar Singh                | High Performance Power Electronic Converter for Solar PV Applications                                                                 | MNNIT Allahabad                  |
| 12. | July 2024         | Jyotirmoy Saharia                  | Isolated DC-DC Converters for EV Application                                                                                          | MNNIT Allahabad                  |
| 13. | July 2025         | Mohammad Asif Ali                  | Improved Performance On-board charger for EV Application                                                                              | MNNIT, Allahabad                 |
| 14. | July 2025         | Aman Keshari                       | Characterization of Wide Bandgap Power Semiconductor Devices for improved Performance and EMI Mitigation                              | MNNIT Allahabad                  |
| 15. | July 2025         | Shikha Barti                       | High Gain Converter based Renewable Energy Integration in DC Microgrid                                                                | MNNIT Allahabad                  |
| 16. | June 2026         | Priya Rani                         | Design, Development and Analysis of High Gain DC-DC Converters for Electric Vehicle Applications                                      | MNNIT, Allahabad                 |
| 17. | June 2026         | Vishruti Singh                     | High Performance Onboard Chargers for EV Applications                                                                                 | MNNIT Allahabad                  |
| 18. | June 2026         | Shoaib Akther                      | SRM Drive with improved performance for Solar Water Pumping Applications                                                              | MNNIT Allahabad                  |

#### Workshops & Seminars

- 1 Participated in Workshop on Advanced Control and computing technologies for reliable Industrial Electricity saving (acctries-2014) from 17-18 Feb. 2014 Organized by Department of Water Resources Development and Management & Indian Water Resources Society (IWRS) at Department of Water Resources Development and Management, IIT Roorkee.
- 2 Attended an AICTE recognized ICT based program 5 days' workshop, "The FDP on soft computing", NITTTR, Chandigarh, 11- 15 November 2013.
- 3 Attended a National Workshop on MATLAB and its Applications" conducted at GMR Institute of Technology in Dec 2008.
- 4 Organized an IEEE associated workshop 'CIRCUIT SAPIENS' on MATLAB and PCB Design for Engineers at Graphic Era University in Mar 2013.
- 5 Delivered a Seminar at IEEE International Conference at MNNIT, Bhopal
- 6 Delivered a Seminar at the IEEE International Conference at PSNA College of Engineering.
- 7 Delivered a Seminar at the IEEE International Conference at PSNA College of Engineering.
- 8 Delivered a Seminar at IEEE International Conference at Nagapatnam.
- 9 Delivered a Seminar at National Conference at MVGR, Vizainagaram.
- 10 Curriculum seminars at IIT Roorkee

#### Personal Details

|                                   |                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------|
| <b>Father's name</b>              | Venkata Rao Dogga                                                                       |
| <b>Date of birth</b>              | 18 <sup>th</sup> April 1988                                                             |
| <b>Gender</b>                     | Male                                                                                    |
| <b>Marital Status</b>             | Married                                                                                 |
| <b>Nationality</b>                | Indian                                                                                  |
| <b>Linguistic Proficiency</b>     | Telugu, English, Hindi                                                                  |
| <b>Permanent address</b>          | 10-82, Srinivasa Colony, Srungavarapukota, Vizianagaram District, Andhra Pradesh-535145 |
| <b>Address for correspondence</b> | IHA-003, Staff Colony, MNNIT Allahabad, Uttar Pradesh 211004                            |

#### References

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prof. M. K. Pathak</b><br>Professor<br>Electrical Engineering Department<br>IIT Roorkee<br><a href="mailto:mukesh.pathak@ee.iitr.ac.in">mukesh.pathak@ee.iitr.ac.in</a><br>+91 9458314144 | <b>Prof. Rached Dhaouadi</b><br>Professor<br>College of Engineering<br>American University of Sharjah<br><a href="mailto:rdhouadhi@aus.edu">rdhouadhi@aus.edu</a><br>+971-505393239 | <b>Mrs. Poojitha Rajana</b><br>Director<br>Zunik Energies (P) Ltd.<br>I-2, TIDES, IIT Roorkee<br><a href="mailto:ceo@zunikenergies.com">ceo@zunikenergies.com</a><br>+91-7989308266 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

I hereby declare that all the information provided above by me is correct to the best of my knowledge.

Date: 25/10/2024

Place: Prayagraj, India

Dr. Dogga Raveendhra